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AstroGate: A conceptual design study for a post-ISS commercial crewed space station

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Abstract

The largest space structure ever to be placed in space, the International Space Station (ISS), has been revolutionary for research and has been home to many astronauts. As the ISS is aging rapidly and the plans for its retirement are already in motion, the need for a modern, sustainable, and state-of-the-art space station that drives research and prepares humanity to explore further is already rising. To address this need, a conceptual design study resulting in 'AstroGate' was conducted at the Space Station Design Workshop 2023 by 19 international interdisciplinary students and young professionals under the guidance of 20+ subject matter experts at the Institute of Space Systems of the University of Stuttgart, Germany.

AstroGate is a comfortable and modular space station with commercial production capabilities for unique products such as high-performance optical fibers, 3D printed organs with high demand on Earth, and an expanding large-scale manufacturing facility to enable a new era of space exploration with Mega Components. With the launch of the minimum viable station in 2030, AstroGate will be constructed and launched in stages with its modular approach and fully operational in 2039 with a capacity of hosting ten astronauts. With a commercial yet innovative approach, AstroGate will have a hybrid life support system, advanced robotic capabilities, an artificial gravity habitation area, and a manufacturing platform for large-scale structures. This capability will help humanity overcome the size and scale of components required to enable missions into the deep unknown beyond our solar system.

This design study includes the design of all subsystems, including structures, propulsion, life-support, communication, radiation, power, and thermal. Most of the proposed technologies have high levels of technology readiness, making the concept highly realizable. Taking into account the existing market, a cost estimate is prepared for the entire project with the launch plan. The risks associated with the project and their mitigation strategies are also discussed. A business plan includes commercial use cases, promotion, and collaboration strategies.
Keywords: Space Station, In-Space Manufacturing.

Acronyms/Abbreviations

AOCS Attitude and Orbit and Control System
ECLSS Environmental Control and Life Support System
EMU Extravehicular Mobility Units
EPS Electrical Power System
ESA European Space Agency
FACT Flexible Advanced Concentrator Technology
GCR Galactic Cosmic Ray
HPGP High-Performance Green Propellant
ISMA In-Space Manufacturing and Assembly
ISS International Space Station
ITAR International Traffic in Arms Regulations
LEO Low Earth Orbit
MBS Mobile Base Systems
MVP Minimum Viable Product
PMAD Power Management and Distribution
ROSA Roll-Out Solar Array
SPDM Special Purpose Dexterous Manipulator

SPE Solar Proton Event
TCS Thermal Control System
TRL Technological Readiness Level

1. Introduction

In 2023 at the Institute of Space Systems in Stuttgart, Germany, two teams each consisting of 20 students and young professionals from around the world were challenged to develop a space station for a post-ISS era. With equal resources and experts available for guidance in the development process, the Red team presented the concept of 'AstroGate' as shown in Figure 1 which was voted as the superior concept leading the team to win the competition at this flagship one-week intensive workshop. The

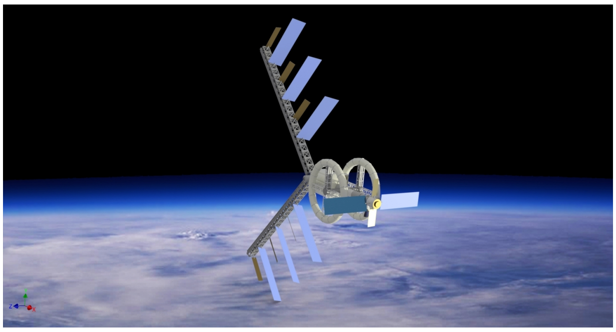

Fig. 1. The AstroGate Space Station

mission was to research and conceptualize a comprehensive study of a post-ISS crewed platform in the Earth orbit. The proposed concept of the crewed platform shall enable a permanent base as a successor of the ISS providing a test bed for scientific and commercial applications. Hence, the concept should provide the infrastructure for large-scale, mainly fully automated manufacturing and assembly capabilities enabling space-based megastructures supervised by a permanent crew of at least 10 astronauts. The AstroGate was designed with primary and secondary objectives fulfilling the above-mentioned mission with the following vision for the station: "AstroGate is the comfortable space gate with commercial production capabilities for unique products with high demand on earth and a large-scale manufacturing facility to enable a new era of space exploration with Mega Components"

2. Concept of Operations

The orbit of AstroGate will be circular (eccentricity = 0) and located in Low Earth Orbit (LEO) at 600 km with an inclination i of 46 °.

The orbit altitude has been selected in order to mitigate the risk of collisions with space debris, which has a higher distribution at superior altitudes, as evidenced by data extracted from the ESA-MASTER tool [1]. Furthermore, the selected altitude is based on a trade-off between a higher ΔV demand and a higher exposure to radiation (which would ultimately lead to an increasing mass due to higher protection measures) at higher altitudes, while considering the mitigation of the orbital decay due to atmospheric drag at lower altitudes, taking into account the mission lifetime and transportation costs. Moreover, the circularity of the orbit is of significant importance concerning the complexity of necessary frequent rendezvous and docking scenarios that are anticipated to occur during the mission. An orbit with an excessive eccentricity would unnecessarily increase the required ΔV, necessitating more complex orbit maneuvers for each transportation mission.

The choice of a circular orbit does not affect the complexity of the mission scenario concerning dynamic factors such as thermal loads or illumination/eclipse time. The orbital inclination has been selected to be 46° to facilitate the use of the most prevalent and accessible launch sites, including the Baikonur Cosmodrome in the future or potential emerging launch sites. This provides redundancy if the currently included launch sites become unavailable for reasons beyond the mission design, whether due to unforeseen technical, environmental, or political factors such as Baikonur Cosmodrome nowadays. Furthermore, there are no compelling reasons for having an inclination of AstroGate's orbit higher than 46°, as increasing inclinations would unnecessarily lead to greater radiation exposure and a higher ΔV demand.

Given the typical ground track of an object located in LEO, such as AstroGate, it is important to emphasize that only a limited number of ground stations are capable of maintaining contact for a restricted duration based on conducted coverage and visibility analysis. Consequently, as exposed in section *4.2.7 Communications* the continuous contact would be ensured through additional connections to relay satellites, adhering to the safety precautions established for human spaceflight. Various operational modes will be enacted to ensure the safe operation of AstroGate during specific scenarios, including Nominal Mode, Docking Mode, Assembly Mode, Maintenance Mode, Research Mode, ISMA Mode, Emergency Mode, Standby Mode, and Training Mode.

Safety is of paramount importance in the operation of the station and in safeguarding the well-being of the crew. Emergency procedures must be developed and rehearsed to address various contingencies. Among these, for any given crew capacity, there must be a return capsule capable of accommodating the entire crew and ensuring their safe return to Earth.

Regarding the end-of-life phase of AstroGate, anticipated in the latter half of the 21st century, it is imperative to reduce the station's orbit to a sufficiently low altitude to enable atmospheric drag to facilitate its reentry and disintegration within the atmosphere. To commence the deorbiting procedure, a ΔV of 280 m/s is required for the terminal orbital adjustment. It is crucial to note that the total mass of the fully assembled AstroGate necessitates a propulsion system that is currently unavailable, such as the U.S. Deorbit Vehicle, which SpaceX is developing for NASA, to perform the intended deorbit maneuver ISS.

3. Business Model

AstroGate's strategy has short-term and long-term goals. Short term: produce unique products for Earth. Long term: maintain production cash flow and enable the next era of space exploration by providing Mega Components, turning visionary missions into reality.

3.1 Short Term

In the short term, the AstroGate station will provide a pressurized workshop equipped with diverse manufacturing apparatus for commercial enterprises. These enterprises can leverage the unique microgravity environment to manufacture innovative products exclusive to space. The material feedstock will be launched from Earth, transformed into valuable products, and subsequently returned to Earth for commercialization. Given the station's design as an adaptive entity, AstroGate can accommodate new customer requirements, enhance the manufacturing workshop's capabilities, or even incorporate entire production lines. Generally, the AstroGate workshop is versatile in addressing various needs and is receptive to innovative product ideas and business models. Certain products significantly benefit from the microgravity environment, while others can only be produced under such conditions. For these scenarios, AstroGate will prove to be exceptionally beneficial. The notable such products are:

3.1.1 High-performance optical wires

Producing high-performance optical fibers in space's microgravity is an advanced and promising application. ZBLAN, with much lower signal losses than silica-based fibers, can only be made in microgravity to avoid gravity-induced impurities. This reduction in signal loss eliminates the need for repeaters, reducing latency, which is crucial for financial markets. NASA projects an annual profit of USD 2.5 billion, including expenses [2]. Companies like Made in Space and Physical Optics Corporation have tested ZBLAN production on the ISS. Customers can send inexpensive raw materials to AstroGate, where fibers are made in the glass alloy manufacturing apparatus. On Earth, one kilogram of ZBLAN fiber (30 km in length) sells for USD 6 million. In-space ZBLAN fiber production is the most mature and profitable business case, making it an ideal initial venture.

3.1.2 3D-Printed Organs

The Global Observatory on Donation and Transplantation estimates that approximately over 150,000 organ transplants occur annually [3], yet this only addresses less than 10% of the global transplant demand. Researchers have experienced limited success with 3D printing human tissues on Earth [4]. Fabricating soft human tissues, such as blood vessels and muscles, is particularly challenging due to gravitational forces. When using smooth, easily flowing biomaterials that closely mimic the body's natural environment, the tissues collapse under their weight, resulting in ineffective constructs. However, in a microgravity environment, such as that provided in space, 3D-printed soft tissues can maintain their structural integrity. Nonetheless, without proper post-processing, space-printed tissues would collapse upon immediate return to Earth. To address this, the BioFabrication Facility, developed by Redwire [5], incorporates a cell-culturing system that gradually strengthens the tissues over time, rendering them viable and structurally self-supporting once reintroduced to Earth's gravitational conditions.

3.1.3 Pharmaceuticals

A McKinsey analysis report [6] identified pharmaceuticals as a promising industry for microgravity applications. Pharmaceutical research in microgravity shows promise, as more potent drugs can be produced in zero-g. Protein manipulation works better in weightlessness. The analysis demonstrates an estimated potential annual profit of USD 2.8 to USD 4.2 billion in the short to medium term. Varda Space recently also launched its first satellite to produce pharmaceuticals in orbit, with the aim to make a business model demonstration.

3.2 Long Term

AstroGate will operate an extensive in-space manufacturing facility to advance the frontiers of space exploration. Presently, there is no provider for large-scale space components. Nonetheless, a substantial demand exists, as numerous components would benefit significantly from increased dimensions. AstroGate aims to address this requirement by supplying Mega Components. These Mega Components are focused on dimensions rather than mass. AstroGate's business model does not initially depend on space resources, although they can be incorporated in the future. Prospective customers include major space agencies and emerging commercial entities.

3.2.1 Giant Solar Arrays

A large-scale, megawatt-level, lightweight solar array represents a viable solution for missions necessitating higher power for deep space exploration. The availability of power is a significant constraint on electric propulsion.

With an adequate energy supply, it becomes feasible to develop new spacecraft with substantially increased budgets (while maintaining proximity to the sun). This advancement would enable significantly faster missions to Mars, facilitate lunar space tugs, and even support asteroid mining operations.

3.2.2 Mega Telescopes

The James Webb Space Telescope, with its 6.5-meter mirror, is revolutionizing our view of the universe by capturing detailed images of distant galaxies and exoplanets. The upcoming LUVOIR telescope, designed with a mirror up to 15 meters in diameter [7], promises even more groundbreaking discoveries. With its ability to observe a wider range of wavelengths and detect faint objects, LUVOIR could help identify potentially habitable planets and uncover new cosmic phenomena. As we scale up to even larger space telescopes, reaching diameters of hundreds of meters, the insights gained could redefine our understanding of the cosmos on an unprecedented level.

3.2.3 Planetary Sunshades

Climate change represents one of the most significant challenges faced by humanity today. A planetary sunshade presents a potential space-based solution or at least a means of mitigation. Due to the extensive shading area required, a sunshade could serve as a long-term project and consistent client for numerous years.

Considering the short-term and long-term opportunities to utilize a large and technologically advanced platform, AstroGate presents a lucrative business case for producing unique products and Mega Components. Additional possibilities, such as studying human physiology in microgravity and partial artificial gravity, testing new technologies aboard the station, and expanding research facilities, position AstroGate as a space station designed for large-scale operations.

4. Space Station

4.1 Structure and design

The structure of the AstroGate space station integrates critical considerations for radiation protection, a significant concern for long-term human habitation in space. Space radiation, primarily from Galactic Cosmic Ray (GCR) and Solar Proton Event (SPE), carries great risks to the health of the crew and the integrity of the material. Therefore, the structural design of AstroGate includes advanced radiation protection strategies, leveraging material science and module placement to mitigate these hazards while maintaining cost efficiency.

4.1.1 Structural Architecture and Module Selection

The modular architecture of AstroGate is highly dependent on designs derived from ISS or other developments by private companies, as shown in Table 1, significantly reducing development and manufacturing costs. The use of established module designs such as the nodes, habitats or airlocks streamlines the engineering process and allows the station to benefit from years of operational experience. This heritage-based design enables AstroGate to avoid the significant research and development costs associated with entirely new modules.

AstroGate module	Similarity
Nodes 1, 2 and 3	ISS Tranquility
Habitat 1 and 2	ISS Leonardo
Laboratory	ISS Destiny
Inflatable Workshop	LIFE 500 [8]
Airlock 1, 2 and 3	ISS Quest
Centrifuge A and B	Nautilus-X [9]

Table 1. AstroGate modules similarity with existence developments

The primary structure of AstroGate is built around lightweight but strong aluminum-lithium alloys that provide the necessary mechanical strength while minimizing launch mass (see Table 2). The external trusses, constructed from graphite-epoxy composites, support large solar arrays, radiators, and scientific instruments. These trusses also house propulsion systems, robotic arms, and docking systems, integrating with other subsystems while ensuring overall structural stability.

Module	Length (m)	Diameter (m)	Mass (kg)
Node 1, 2 & 3	6.7	4.48	19,000
Hab 1 & 2	6.6	4.57	10,000
Laboratory	8.4	4.2	14,515
Workshop	13 (inflated)	15 (inflated)	11,000
Centrifuges	30.0 (OD)	26.5 (ID)	10,000
Trusses	100.0	4.9	20,000

Table 2. AstroGate module properties

Using designs derived from ISS, AstroGate not only benefits from established reliability, but also achieves significant cost reductions in both design and construction. These cost savings are further enhanced by standardizing interfaces, docking ports, and structural components, allowing seamless integration of future modules.

4.1.2 Micrometeoroid and Orbital Debris Protection

In addition to radiation, protection of micrometeorites and debris is another structural challenge for long-term

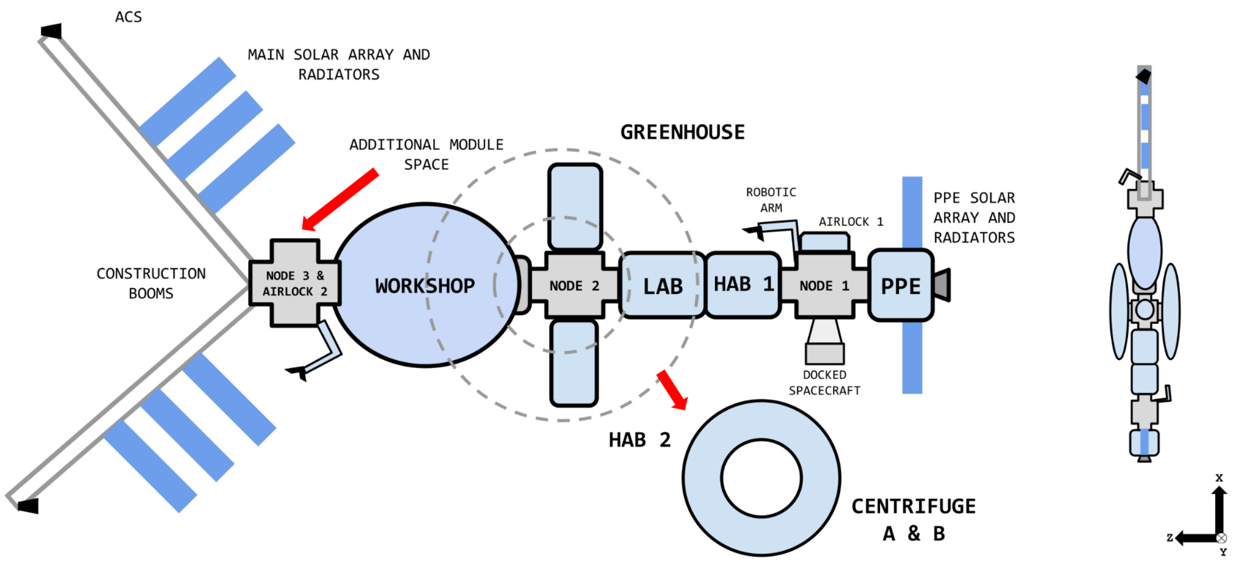

Fig. 2. AstroGate space station layout baseline concept (not to scale)

space operations. AstroGate employs Whipple shields, a proven method to protect spacecraft from high-velocity impacts. These shields consist of multiple layers of thin materials, such as aluminum and kevlar composites, designed to absorb and disperse the energy of incoming debris before it reaches critical systems.

The external trusses, which support solar arrays and radiators, are particularly vulnerable to debris impacts because of their size and exposure. As such, they are constructed from graphite-epoxy composites, which are lightweight yet capable of withstanding both micrometeoroid impacts and the forces exerted during station operations, such as attitude adjustments or orbital maneuvers. These trusses are the key to ensuring the station's overall stability while reducing the risk of damage from orbital debris.

4.1.3 Subsystem Integration and Structural Considerations

The AstroGate structure is designed to support and integrate with a variety of critical subsystems, ensuring long-term functionality of the station. The Attitude and Orbit and Control System (AOCS) and propulsion systems are mounted on external trusses, designed to handle the forces exerted during station maintenance and docking operations. The solar arrays of the station on the trusses must be aligned to provide optimal power generation while minimizing structural stress.

AstroGate's robotic systems, which include mobile base systems and robotic arms, are also integrated into the external trusses. These arms are used for module assembly, maintenance, and extra-vehicular activities. The trusses are designed with attachment points and rails, allowing the robotic systems to traverse the station as needed and providing flexibility for operations.

One of AstroGate's most novel structural components is the inclusion of dual counterrotating centrifuges. These modules, designed to simulate artificial gravity, are mounted on long booms that extend from the core structure. To avoid transmitting rotational forces and vibrations, these centrifuges must remain isolated from the rest of the station. The carbon fibre-reinforced booms that support these centrifuges are designed to handle these unique stresses while maintaining structural integrity.

4.1.4 Crew Habitability and Ergonomics

In addition to its structural robustness, the AstroGate space station has been designed with habitability of the crew and ergonomic considerations at the forefront. Long-term space habitation requires careful attention to the physical and psychological well-being of astronauts. The design incorporates features that promote a comfortable and efficient living environment for the crew, such as soundproofing to reduce noise, adjustable lighting to regulate circadian rhythms, and ergonomic furniture to prevent musculoskeletal issues.

Each astronaut has access to private sleeping quarters inside Habitats 1 and 2. These cabins are designed to reduce noise and light intrusion, promoting better sleep in the space's 24-hour environment. They are equipped with acoustic damping materials and adjustable lighting systems that mimic the natural day-night cycle, helping regulate astronauts' circadian rhythms and reduce fatigue. Habitats also feature common areas, where astronauts can gather for meals, recreation, and social activities. These areas are designed with adjustable lighting, allowing crew members to create a comfortable environment suited to different activities. Strategically placed windows throughout the station provide views of Earth, offering astronauts a meaningful psychological connection to home.

Centrifuge modules improve habitability by providing artificial gravity for long-term health benefits. Astronauts will have access to private quarters within the centrifuge modules, which simulate gravity environments of up to 0.5 g. Consequently, the physical effects of prolonged exposure to microgravity, such as muscle atrophy and loss of bone density, are expected to be reduced as exposed in the *4.2.5 Environmental Control and Life Support System* section, ensuring that crew members remain healthy during extended missions.

4.2 Subsystems

4.2.1 AOCS and propulsion

The AstroGate station's primary propulsion system, operating in tandem with the AOCS, will handle orbital maneuvers and altitude adjustments to ensure precise alignment with mission requirements. As detailed in the *7 Implementation and Schedule* section, the first module to be deployed to orbit will be the Power Module. Although this module lacks an independent propulsion system, it will feature cold-gas thrusters to manage attitude and orbital control. Although the collision probability during this phase is low, these thrusters may be used to avoid collisions if necessary, ensuring additional redundancy in the early assembly phase. Within a week of the power module launch, an upgraded and redesigned SpaceX Cargo Dragon spacecraft will follow, docking with the power module and temporarily functioning as the station's propulsion unit. The Power Module contains a dedicated propellant tank with a capacity of 422 kg and a volume of 0.36 m^3. This fuel reserve is designed to facilitate orbital maintenance and collision avoidance maneuvers, providing a ΔV capacity of 3.45 m/s annually without relying on the propellant of the Dragon spacecraft. The propellant tank will be refueled annually, incorporating a safety margin 20% to accommodate unforeseen events and operational fluctuations.

An advanced cargo spacecraft with next-generation propulsion will dock with the power module to enhance the station's propulsion by increasing thrust, efficiency, and maneuverability. This flexibility is crucial for diverse mission objectives. The early implementation of an independent propulsion system would constrain mass, power, and thermal management, key factors in manned missions. Using visiting cargo spacecraft for propulsion reduces these demands, eliminating the need for additional power and thermal systems. Coordination between resupply missions and propulsion operations is part of the mission timeline for consistency, as shown in Figure 7.

The station orientation system is meticulously engineered to mitigate momentum disturbances. This is achieved by counterrotating the centrifuges, neutralizing their angular momentum, and minimizing the impact on the overall stability of the station. The AOCS incorporates central moment gyroscopes, chemical thrusters, reaction engines, and cold gas thrusters. The cold gas thrusters will be critical in the early assembly phase to stabilize the propulsion module in orbit before the arrival of subsequent modules. Additionally, 12 High-Performance Green Propellant (HPGP) LMP103s [10] from Bradford Ecaps will control attitude during dynamic operations, such as spacecraft docking, and provide recovery when the gyroscopes reach their operational limits. 6 thrusters will be mounted on each truss member to ensure full 3-axis control. Initially placed on Node 2, these thrusters will be relocated to the rear trusses using robotic arms once the station assembly is complete. Redundancy is ensured by the choice of 6 thrusters per truss.

To support an annual impulse time of 3000 seconds, 4003 kg of propellant will be stored in a 2.2 m^3 volume tank for the 12 thrusters. The selection of HPGP over hydrazine offers improved performance [11] due to the higher density of HPGP, which allows more efficient storage in a smaller tank with structural benefits. Importantly, HPGP aligns with crew safety objectives due to its lower toxicity and non-carcinogenic properties, reducing the ground processing time before launch. If no cargo spacecraft is docked, the station's AOCS thrusters can be used for orbital maintenance. A 20% propellant margin has been added for AOCS thrusters to improve redundancy. For collision avoidance, upgraded SpaceX Cargo Dragon or Cygnus spacecraft thrusters will fire for 10 minutes, providing 880N of thrust and raising the orbit by 4km.

To ensure safe deorbiting of the station upon the conclusion of its mission, reliance will be placed on advanced cargo spacecraft, such as the U.S. Deorbit Vehicle previously mentioned or, more likely, SpaceX's Starship.

4.2.2 Energy

The Electrical Power System (EPS) must reliably supply power during all mission phases, including eclipses, while being scalable to accommodate future station expansions. It should provide redundancies to mitigate component failures and facilitate the easy replacement of life-limited parts. Crew safety and protection of Earth's population and environment at the station's disposal are also critical considerations.

The power demand during different stages of station assembly (see section *7 Implementation and Schedule*) is shown in Table 3. As EPS expands alongside the station, the system must be able to meet the power requirements during each intermediate stage. The phase Minimum Viable Product (MVP) extends until the first truss elements are installed, while Phase 2 includes the installation of additional external manufacturing truss segments. In addition, it needs to be scalable to support future station expansion, provide redundancies in case of component failures, and allow replacement of life-limited components.

Subsystem	MVP	Phase 2	Final
TCS	2	11	13
ECLSS	13	26	39
Greenhouse	0	42	42
Communications	1	1	1
Propulsion	2	3	3
ISMA	8	90	100
Robotics	1	1	3
Centrifuge	0	0	5
Miscellaneous	4	7	16
Total	**31**	**181**	**222**

Table 3. Average Modules Power Demand [kW]

Various options for the EPS architecture were considered. Nuclear fission reactors were dismissed due to concerns about immature technology, crew radiation risks, legal challenges, and environmental impact during station decommissioning. A solar dynamic system with thermal eclipse storage was considered due to its reduced drag [12], but ultimately rejected due to its higher mass and low Technological Readiness Level (TRL). Roll-out solar arrays were selected as the optimal solution because of their high TRL, space tested at ISS, low cost, safety, and modularity. Lithium-ion batteries were preferred for energy storage over regenerative fuel cells because they offer better round-trip efficiency and a proven track record.

The main components and energy flows are shown in Figure 3. The MVP station is powered by two 8x25 m Redwire Roll-Out Solar Arrays (ROSAs) [13]. The capacity for the later phases is distributed across the truss. In addition to batteries and support equipment, each segment of the truss contains a 9× hinged two-axis 39 m ROSA, enhanced with Redwire Flexible Advanced Concentrator Technology (FACT) [14]. This allows for modular expansion as needed. A 20-year lifetime at 2.75 % annual degradation is targeted before replacing solar arrays. Batteries have a 30-year lifespan. To ensure long-term reliability, the power generation and storage capacity are first sized according to the end-of-life parameters, considering the power demand and orbital data (Table 3). The beginning-of-life sizes are then calculated to account for the expected degradation of solar arrays and batteries over time. The power-to-mass ratio of the solar array is estimated at $175\,\mathrm{W\,kg^{-1}}$, reflecting a 33 % improvement over previous designs, driven by advances in technology and scalability, alongside an additional 10 % performance boost from FACT technology [14]."

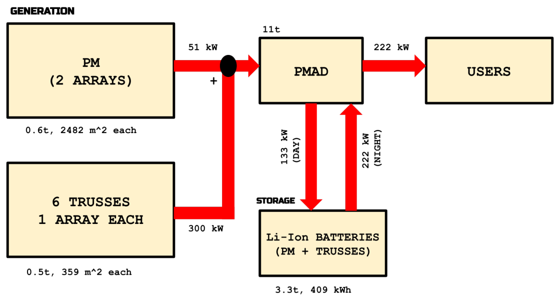

Fig. 3. AstroGate main energy flows

The battery energy density is set at $180\,\mathrm{W\,h\,kg^{-1}}$ [15], factoring in packaging constraints. Estimating the mass of Power Management and Distribution (PMAD) system is more challenging without a detailed design, but a value of $50\,\mathrm{kg\,kW^{-1}}$ (average power) is assumed [16], considering anticipated technological advances and the large dimensions of the station. This results in a total power system mass of 19 tonnes for a total power capacity of 222 kW.

4.2.3 Radiation

The radiation protection subsystem on the AstroGate space station mitigates long-term health risks posed by space radiation, specifically GCRs and SPEs. The system utilizes passive shielding that involves high-efficiency materials in conjunction with potential active shielding approaches under exploration. The design is guided by stringent requirements for astronaut health and operational safety during extended space missions.

Reliable real-time radiation monitoring is integral to the subsystem. AstroGate employs thermoluminescent dosimeters and optically stimulated luminescence dosimeters distributed throughout the spacecraft and worn by crew members to track radiation dose accumulation. These dosimeters measure the absorbed dose of ionizing radiation and alert the crew when thresholds are exceeded, enabling real-time risk mitigation. Accurate ground-based calibration ensures consistency and reliability in the measurements, which is critical to maintaining acceptable risk levels in interplanetary missions.

The selection of material for space shielding is complex and requires an optimal balance between radiation protection, mass constraints, and secondary radiation interaction. Polyethylene was selected for AstroGate because of its high hydrogen content, which interacts favorably with high-energy particles such as protons and heavy ions. The hydrogen atoms effectively scatter and absorb radiation, significantly reducing penetration. Studies have shown that a 7 cm polyethylene thickness offers substantial protection while minimizing spacecraft mass, a crucial factor in space mission planning.

In contrast, aluminum, while commonly used for spacecraft structures, is less effective in deep space environments due to the potential to generate secondary radiation when struck by high-energy particles. Polyethylene reduces this risk and performs better in terms of minimizing secondary radiation exposure. This material is particularly effective against GCR, one of the most dangerous forms of space radiation for long-duration missions.

Active shielding methods are also being explored to complement or potentially replace passive shielding. An approach involves using magnetic fields to deflect charged particles. Superconducting magnets could generate these fields around the spacecraft, offering dynamic protection without adding significant mass. However, producing the required field strengths (10–20 teslas) presents technical challenges, such as high power consumption and the potential adverse effects on the crew from prolonged exposure to intense magnetic fields. An alternative to magnetic shielding involves electrostatic fields, which repel positively charged particles such as protons. However, maintaining charge balance in the spacecraft's environment, particularly in deep space, where electrons can neutralize these fields, remains a significant hurdle. Hybrid shielding designs that combine magnetic and electrostatic fields are also under investigation. Although they are still in the theoretical phase, these hybrid systems could reduce power requirements while maintaining effective radiation protection.

NASA's radiation exposure limits are set at a maximum of 3% risk of exposure-induced death from fatal cancers throughout an astronaut's career [17]. AstroGate's radiation protection subsystem is designed to maintain crew radiation exposure well below this threshold, even during extended missions. Short-term dose limits are also imposed to prevent acute non-cancer effects such as nausea, fatigue, or organ damage from high doses of radiation.

To mitigate the effects of SPE, which can increase radiation flux to lethal levels within hours, AstroGate incorporates storm shelters. These are small, highly shielded areas within the spacecraft where the crew can seek refuge during such events. These shelters use additional layers of polyethylene and materials such as water, which can be strategically placed around the crew area to improve radiation protection. In addition, wearable radiation shielding is being considered. These lightweight protective suits, made from hydrogen-rich materials such as polyethylene, would provide personal protection, especially during activities where localized radiation exposure might exceed safe levels.

Radiation simulations confirm that passive material shielding and potential active methods significantly reduce crew exposure. For example, during solar maximum periods, simulations estimate a 0.661% risk of cancer due to radiation exposure, well below NASA's career REID limit [17]. These results confirm the system's effectiveness in maintaining radiation exposure within safe limits for long-duration missions, ensuring compliance with the established safety standards for space exploration.

4.2.4 Thermal Control System (TCS)

The TCS for the AstroGate space station ensures that all spacecraft components remain within predefined operational temperature ranges under various mission conditions, including safe mode and artificial gravity. The TCS consists of passive and active systems, each fulfilling specific roles.

The passive thermal control system maintains temperatures without the need for powered equipment, making it cost-effective, lightweight, and low-risk. Its key components include:

- **Surface Coatings:** The radiators are coated with fluorinated ethylene Propylene tapes, which reflect solar energy while efficiently emitting thermal energy.
- **Module Insulation:** Multilayer insulation minimizes radiation heat loss, comprising Kapton and polyimide films.
- **Bolted Joint Insulation:** Thermal insulating bolts manage temperature gradients between materials with different thermal conductances.
- **Debris Shield:** Integrated into the insulation to provide additional protection.

Passive systems handle solar and Earth radiation, while the active control system primarily manages internal heat loads.

Active thermal control systems are dependent on electrical power and are essential for regulating the temperature of high-temperature load components. The AstroGate utilizes both single-phase (SP) and dual-phase (DP) fluid loops for this purpose (see Table 4). These systems allow for precise temperature control, particularly for sensitive equipment.

AstroGate uses rotatable radiators, designed to dissipate excess heat into space. These radiators are supported by single-phase and dual-phase cooling loops depicted in Figure 4, with dual-phase loops offering 50% more efficiency. The cooling loop system is highly redundant, allowing loops to be bridged as necessary to reduce power consumption or transfer heat between modules. This capability ensures that, even in the event of failure, heat management can continue efficiently. Thermal calculations show that in the hot case (when the Sun-facing side of the station is exposed to sunlight), the equilibrium temperature can reach around 370 K. In contrast, during the

Loop	Component Serviced	Type	Connection
1-2	PPE	SP	Outer loops
3-4	Lab, Hab1, Node1	SP	Outer loops
7-8	Greenhouse, Workshop, Hab2	DP	Redundancy
9-10	CentrifugeA, CentrifugeB	DP	Redundancy
11-12	Manufacturing trusses	SP	Outer loops

Table 4. Cooling loops configuration of the TCS

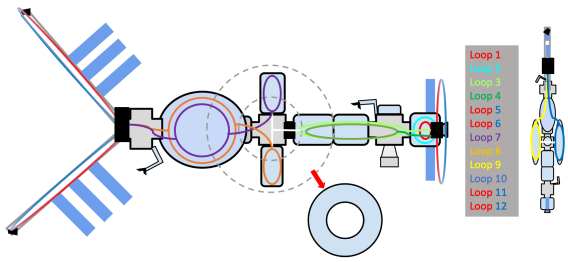

Fig. 4. Layout of fluid loops in the AstroGate TCS

cold case (when the entire station is in shadow), the equilibrium temperature drops to 256 K.

To maintain proper thermal regulation, the system requires approximately 87 m^2 of radiators for the MVP and approximately 433 m^2 of radiators for the entire station, as illustrated in Figure 4. The TCS consumes an estimated 12 kW of power and operates using approximately 750 liters of coolant fluid.

Although passive and active thermal control components are based on proven technologies, the two-phase cooling system represents a novel innovation. This system is currently awaiting final validation, which is expected in 2033.

4.2.5 Environmental Control and Life Support System (ECLSS)

The AstroGate Space Station is designed for a minimum mission duration of 15 years, supporting crew sizes ranging from 4 to 10 astronauts at different assembly stages. The ECLSS and Human Factors are optimized to adapt to the evolving crew size of the station and ensure crew well-being, health, and operational efficiency (see Fig. 5).

Habitat I, launched by 2032, has life support systems similar to ISS, which supports four astronauts using physicochemical technologies. These technologies recycle up to 90% of water and recover 42% of oxygen [18]. Key components include a four-bed Molecular Sieve, a Trace Contaminant Control System, an Oxygen Generation System, and a Vapor Compression Distillation Unit [19]. Resupply missions are required for food provisions.

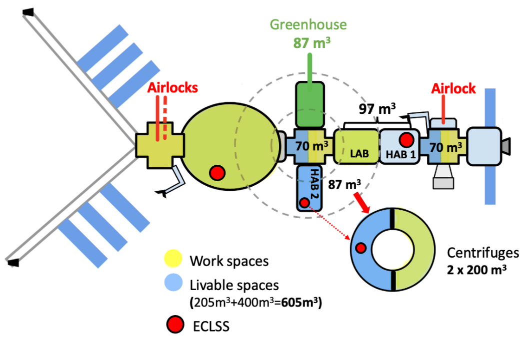

Fig. 5. AstroGate space station layout, highlighting the life support systems

In 2037, Habitat II and the Workshop will replicate Habitat I's life support systems with additional oxygen supplementation from the propulsion subsystem. The condensation water from the air conditioning system will be filtered and reintegrated into the loop through the TCS. This shows the integrated nature of the life support systems in the station. The introduction of the Greenhouse module in 2037 will incorporate a hybrid life support system. A photobioreactor will be used, providing oxygen and food production based on algae. With 40% Spirulina Platensis and 60% Chlorella Vulgaris microalgae. The photobioreactor will cover 15% of the daily nutritional needs of 10 astronauts [18]. The greenhouse, based on the EDEN ISS concept [20], offers 50 m^2 of cultivation space, providing 500 g of fresh vegetables per astronaut daily and contributing to psychological well-being.

In 2039, two centrifuges will be integrated to counteract bone density loss in astronauts, estimated to be 1.0-1.2% monthly. Operating at 4 rpm, these centrifuges will generate 0.5 g on the astronauts' legs, effectively mitigating bone loss [21]. The total mass of the LSS, including emergency and backup systems, is 2300 kg, with 960 kg resupplied every six months. The redundancy of the system components reduces the probability of failure to 0.4%. Power consumption is estimated at 13 kW for each Habitat and Workshop, with the greenhouse requiring 41.8 kW [22].

4.2.6 Robotics

The robotic systems of the station are integral to its operation, enabling efficient maintenance, assembly, and inspection tasks with minimal human intervention. The fully assembled space station will be equipped with the following robotic systems:
- 3× Robotic arms, each with 7 degrees of freedom and touch-sensing technology for enhanced steering capabilities.
- 3× Mobile Base Systems (MBS), each paired with a robotic arm, enabling the smooth transport of In-Space Manufacturing and Assembly (ISMA) parts, cargo, and crew throughout the station.
- 1× Special Purpose Dexterous Manipulator (SPDM) with a minimum of 15 degrees of freedom.
- 2× Humanoid robots will assist the crew with station maintenance, significantly reducing the need for human labor.
- 10 to 40 small AI-driven, four-legged crawler robots for inspection purposes.
- 1× Precision robot dedicated to assembly tasks within a clean room environment.
- 1× Robotic arm with 6 degrees of freedom for object handling inside the clean room environment.

The robotic system will be divided into two operational loops. The first loop will consist of one MBS module, one robotic arm comparable in size to CanadArm2, and one dexterous manipulator unit similar in design to SPDM. This loop will be fully functional within the MVP phase, allowing a robotized system to simplify the station development process. The second loop will operate on the opposite side of the station, assisting ISMA operations and providing similar functionalities. In the early years of operation, MVP modules will require a robotic arm system capable of traversing the entire length of the station.

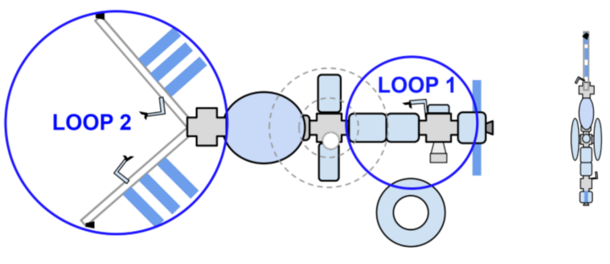

Fig. 6. AstroGate robotic loops location

The MBS module will provide both data and power to its attached robotic arm. In addition to MBS, each module will have two 90-degree slit attachment points on the rail on which MBS will travel. These connections will be made using latching end effectors. The system will allow for docking of new modules, inspections using cameras, and maintenance activities. Given the maturity of these technologies, the station will be fully operational at this stage. The second loop will be established during the solar panel assembly within the ISMA area, with the MBS railing integrated into the trusses to facilitate panel manipulation and handling.

The EVA activities would be supported by robotic arms throughout the station with MBS modules and robotic arms utilizing a mobile foot-resistant device. Non-tethered spacewalks could also be conducted with the utilization of a manned maneuvering unit. For astronaut safety, we would utilize Extravehicular Mobility Units (EMU) space suits similarly as it is on ISS. It is possible to upgrade them as newer technologies are developed.

4.2.7 Communications

The communication subsystem of the AstroGate space station ensures a seamless exchange of telemetry, command, and scientific data between the station, its crew, and ground control. The core of this system is the Ka-Band communication system, which provides high-data-rate transmissions that are critical for supporting routine operations and scientific experiments. The system will use a high-gain steerable antenna and a quadrature phase shift keying modulation scheme to achieve data rates up to 600 Mbps, ensuring efficient transmission of large datasets generated by the station's onboard experiments and commercial activities.

The Ka-Band system, known for its reliability, connects to a geosynchronous satellite constellation, providing uninterrupted communication coverage in all operational modes, nominal, docking, and emergency. This architecture ensures that the station remains in constant contact with Earth-based operations, even during critical mission phases. In parallel, a redundant S-Band system is implemented to take over in case of Ka-Band failure. This built-in redundancy guarantees the safety of the crew and the success of the mission, maintaining communication capabilities even in the event of unexpected failures. AstroGate's ground segment is designed to provide comprehensive global coverage. The primary ground station is settled to be in the US, with the possibility of adding secondary ground stations at Europe and Asia through a partnership agreement. This arrangement ensures minimal latency and reliable communication at all times, including during high-demand operations such as data-heavy scientific research or in-space manufacturing processes.

Based on the station's 600 km orbital altitude, a rigorous link budget analysis confirmed the subsystem's ability to maintain a robust signal-to-noise ratio across all ground stations. The analysis also accounted for a loss in free space of 179.51 dB and an atmospheric loss of 0.03 dB, both of which are crucial factors in maintaining communication reliability during various operational scenarios.

The prospects for the subsystem are promising, with planned enhancements that encompass the incorporation of optical communication technology, such as that made available by Starlink satellites. These advancements will considerably augment the station's data transmission capabilities, especially in the context of impending deep-space missions, where elevated data rates and diminished latency will be imperative.

5. Project Risk Analysis

This analysis focuses on the critical risks that could impact the AstroGate space station project, highlighting those that could disrupt the project timeline and objectives. It is important to note that operational risks, as well as mission-related risks, such as system failures or in-mission incidents, have not been taken into account in this analysis, as the focus remains on the aspects related to project development.

5.1 Launcher Availability

The success of the AstroGate project depends mainly on the availability of Falcon 9 and Falcon Heavy launch vehicles. Any disruption at Kennedy Space Center Launch Complex 39A or Cape Canaveral Space Launch Complex 40 could cause significant delays. It is important to note that Vandenberg Space Force Base, which is capable of launching Falcon 9 rockets, cannot launch AstroGate to the desired target orbit. Alternative launch vehicles, such as New Glenn and Vulcan, also launched from Cape Canaveral, have been identified to mitigate this risk. Finally, the H-IIIA rocket from Japan Aerospace Exploration Agency (JAXA) and Mitsubishi Heavy Industries could be adapted to launch to the desired orbit. Additionally, securing a secondary launch site at the Tanegashima Space Center in Japan provides additional flexibility and reduces the project's vulnerability to delays.

A wide variety of micro-launchers can be used for smaller payloads related to the business model. Preferred options include Rocket Lab's Electron, launching from the Mid-Atlantic Regional Spaceport (MARS) at Wallops Flight Facility, and exploring potential European launch partners operating from the Guiana Space Center (Centre Spatial Guyanais) in Kourou, French Guiana. These alternatives offer additional flexibility and ensure that smaller mission-critical components can be deployed as needed without affecting the overall project timeline.

5.2 Delivery Delays

Delivery delays are a significant concern, which can arise from system integration issues, material qualification challenges, or software development setbacks. To address these risks, rigorous integration testing and simulations are planned to catch subsystem incompatibilities early. Materials will be tested extensively in space-like conditions, with proven alternatives ready as backups. Agile software development practices will be used, with built-in buffer periods to manage potential delays.

A proactive strategy ensures mission critical modules are delivered to the launch site a year in advance, with standby modules available to keep the project on schedule. This means that AstroGate will receive each module one year before launch operations, allowing ample time to address potential delays. Although this contingency measure increases costs, it ensures that the AstroGate project is prepared to handle unexpected setbacks without impacting the overall mission timeline.

5.3 Interface Management Risks

Managing interfaces between subsystems, especially those developed by different teams, presents a substantial risk. Poor coordination could lead to system incompatibilities or integration failures. To mitigate this, precise interface specifications will be established from the start, supported by regular reviews and workshops. A centralized interface management team will oversee the entire process, with partners collaborating to ensure seamless subsystem integration.

6. Cost Analysis

The cost analysis, adjusted to the 2023 costs, of the AstroGate space station project comprehensively examines the financial requirements across its entire lifecycle. It meticulously focuses on critical expenditures in design, manufacturing, launch, and operations. It is important to note that the costs have been calculated based on 2023 figures and have not been adjusted for inflation anticipated in the 2030 decade.

6.1 Critical assumptions

At the core of the financial projections are several critical assumptions that shape the cost estimation process. One of the primary assumptions is that the payment for the initial space station modules will be made to manufacturing partners upon delivery. This results in a significant upfront financial commitment during the early stages of the project. Another foundational assumption is the necessity of maintaining a dedicated mission control center operating 24/7, driven by AstroGate's direct involvement in the launch operations for the first module. This requirement introduces substantial operational costs that are essential for ensuring the safety and success of the mission. Furthermore, the assumption of a two-year astronaut training period before the first mission introduces a critical cost factor that significantly influences the overall budget.

6.2 Cost Breakdown

The $47.564 billion USD financial commitment is distributed across several key areas. The most significant expenditure is attributed to construction and development, amounting to $27.264 billion USD. This cost includes designing, engineering, and assembling the various modules and systems of the station. The strategic reuse of existing designs from the ISS modules and partnerships with established manufacturers have played a key role in reducing these costs. The prelaunch operations, projected at $350.01 million USD, encompass essential preparatory activities prior to the launch of the station. These operations are critical in laying the foundations for the success of the project.

Launch activities constitute another significant financial component, with an estimated cost of $14.749 billion dollars. This category includes the deployment of station modules and other components in orbit. The Falcon 9 and Falcon Heavy launch systems have been particularly effective in reducing launch expenses compared to previous space missions, such as those associated with ISS.

The detailed breakdown of launch costs includes $925 million USD for module launches, $6.603 billion USD for crew mission launches, and $7.222 billion USD for resupply missions. Ongoing operational costs are divided into general operations and ground communications. General operations, which include salaries, training, and ongoing support for ground personnel, are estimated at $5. 200 billion dollars. Communications systems are explicitly budgeted at $790 million USD.

6.3 Cost Efficiency and Strategic Financial Planning

The cost per ton of the AstroGate Space Station is $297.88 million USD, a notable reduction from the cost-specific mass of ISS of $357 million USD per ton. This efficiency is primarily due to the strategic reuse of existing ISS module designs and the cost-saving advantages of the Falcon 9 and Falcon Heavy launch systems. The distribution of costs over the project's 15-year lifecycle is not uniform, with significant financial peaks corresponding to critical milestones. The most substantial expenditure occurs in the initial year, primarily due to the substantial payments for early module deliveries. Subsequent financial peaks align with the delivery of the manufacturing station and the installation of centrifuges, which are essential for the station's operational functionality. Although the average annual expenditure is estimated at $2.85 billion USD, the non-uniform distribution of costs highlights the necessity for meticulous financial planning and resource allocation to manage these peaks effectively.

7. Implementation and Schedule

The AstroGate assembly process will begin in 2030 with the launch of the first propulsion module. The complete construction of the full station is estimated to take 9 years, with a target of becoming fully operational by the year 2039. The diagram in Figure 7 illustrates the deployment strategy, which describes the launch vehicle and cargo arrangements. During the assembly process, the goal is to automate as many tasks as technically possible, employing robotic arms with minimal ground support. The crew will be responsible for supporting the remaining tasks.

Figure 7 illustrates the complete assembly of the AstroGate space station, progressing from the initial stage, to the MVP and finally to a fully operational facility with a capacity for ten crew members. The timeline in Figure 7 depicts the launch of modules using their respective launch vehicles. Resupply missions are highlighted in orange, while crewed missions are shown in blue. The selected launch vehicles for all missions include Falcon Heavy, Falcon 9, and New Glenn, launched exclusively from Kennedy Space Center and Cape Canaveral Space Force Base. Upon a successful test launch, SpaceX's Starship may also be considered for deploying the later modules of the station. Additional launch vehicles and alternative launch sites are detailed in the risk analysis in the *5.1 Launcher Availability* section.

8. Legal and Regulation

In the context of international space law, AstroGate's decision-making process for establishing the space station company headquarters involves careful consideration of existing legal frameworks and Saudi Arabia's Space Activities Law (2018), provide robust governmental support for space sector development despite the lack of established space industries. Europe, with France, Germany, Italy and the United Kingdom as space leaders, has European Space Agency (ESA) support with a mature space sector and is refining its national space laws. The U.S. distinguishes itself with a well-established national space law, significant technological leadership, and active involvement in international agreements like the Artemis Accords. Despite the complexities introduced by International Traffic in Arms Regulations (ITAR) [23], which oversee the export of defense-related technologies, the United States provides a comprehensive legal environment and a robust financial market, making it an attractive location for AstroGate's headquarters.

Legal and regulatory frameworks, international treaties, access to capital, and political stability influence AstroGate's decision. Although the UAE and Saudi Arabia offer strong government support and emerging industries, and Europe has a well-developed space sector, the US was ultimately chosen.

The decision was primarily driven by the extensive legal infrastructure in the US and the vibrant financial market, which outweighed the challenges posed by ITAR regulations. To address these challenges, AstroGate plans to partner with European firms with manufacturing and

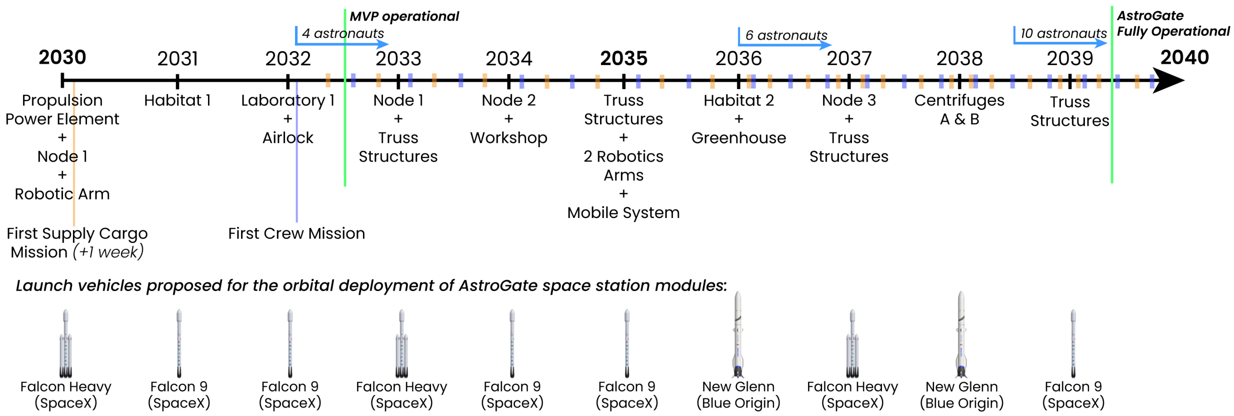

Fig. 7. Development timeline of AstroGate station assembly, crew, and cargo missions.

ITAR compliance experience, ensuring efficient operations. This strategic approach balances the strengths of a robust legal framework and technological leadership with the regulatory complexities, positioning AstroGate for success in its space station project.

9. Impact

9.1 Commercial Impact

In light of the customer demand for a continuity module that extends beyond the projected 2030 lifetime of the ISS, AstroGate has identified two key areas for its strategic focus:

- **Spearheading Cutting Edge In-Space Manufacturing:** Another crucial aspect of AstroGate's strategy is to lead the frontier of manufacturing in space. We understand the immense potential of in-space manufacturing capabilities and are committed to providing the necessary infrastructure and technology for manufacturing cutting-edge products beyond the confines of Earth. Using the benefits of the space environment, we aim to revolutionize manufacturing processes and contribute to the growth of various industries.

- **Enabling Scientific Experiments for Research and Educational Institutions:** AstroGate is dedicated to playing a pivotal role in the future space economy and driving advances in deep space exploration. One of our primary objectives is to serve as an innovative testbed for scientific experiments within our modules. By providing state-of-the-art facilities and resources, AstroGate aims to support educational institutions and researchers in conducting cutting-edge experiments in the unique environment of space.

Through these strategic initiatives, AstroGate aims to solidify its position as a prominent player in the space industry, contributing significantly to the advancement of space exploration and driving innovation in both scientific research and manufacturing.

9.2 Social Impact

AstroGate will have the ability to produce unique products that can only be manufactured in the space environment, handle and manufacture large-scale structures, and artificial gravity modules for use in habitation. These capabilities will drive scientific and industrial innovation that will affect the quality of life on Earth. Advancements in space exploration, life science, robotics, and material science have a positive impact on society.

AstroGate being a commercial company will also enable international partnerships from both national and commercial sides fostering cooperation and diplomacy between different countries and industrial players. The new branches of research driven by the artificial gravity on the AstroGate will pave the way to prepare humanity for future missions in deep space and help us understand more about the human body. Such a space station will increase employment opportunities in various sectors in space and on Earth. It can also stimulate private investment in space-related ventures, leading to economic growth

9.3 Environmental Impact

Considering the much larger size of AstroGate than the ISS, responsible management of space debris will become critical to ensure the sustainability of all space activities. The operation of AstroGate would require energy and resources both on Earth and in space. Minimizing resource consumption by using reusable launch vehicles will be essential to mitigate the environmental impact.

10. Conclusion

AstroGate represents a pivotal advancement in the progression of space infrastructure, conceived not only to succeed the ISS but also to pave the way for future endeavors in space exploration and commercial space activities. Its modular design, alongside state-of-the-art technologies such as artificial gravity, hybrid life support systems, and advanced robotics, positions AstroGate as a highly adaptable platform capable of supporting long-duration crewed missions, as well as substantial manufacturing operations in orbit.

AstroGate's commercial prospects are considerable, with the production of high-demand items such as ZBLAN optical fibers and 3D-printed organs, while its scalable architecture ensures flexibility to future market conditions and technological progress.

As a fundamental enabler of deep-space exploration, AstroGate will be instrumental in the upcoming era of human spaceflight, furnishing the infrastructure necessary for large-scale components and cutting-edge scientific research. By integrating scientific rigor with a viable business model, AstroGate forges a new frontier in space commercialization and exploration. It is not merely a successor to the ISS, but a gateway to the future of space, facilitating humanity to transcend the current boundaries of what is feasible in low-Earth orbit and beyond.

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